

## **Arithmetic Expression Evaluator**

### **Test Case**

**Version 1.0**

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| Arithmetic Expression evaluator | Version: 1.0     |
| Test Case                       | Date: 05/05/2026 |
| EECS348-G10-TC                  |                  |

## Revision History

| Date       | Version | Description                    | Author          |
|------------|---------|--------------------------------|-----------------|
| 05/05/2026 | 1.0     | Section 1,2,3,4,5,6,7 complete | Ayushma Koirala |
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## Test Case

### 1. Purpose

This Test Case Specification document defines a test case for the Arithmetic Expression Evaluator that should be tested. The test cases confirm that the system correctly accepts arithmetic expressions, tokenizes and parses them, evaluates them according to PEMDAS and associativity rules, and returns either the correct numeric result or a clear error message for invalid input.

The tests cover valid integer and floating-point expressions, supported operators, unary operators, parentheses, exponentiation, modulo, and expected error handling behavior such as invalid characters, missing operands, mismatched parentheses, and division by zero.

### 2-5. Test case identifier, Test item, Input specification, Output specifications

The following table lists the main test cases for the Arithmetic Expression Evaluator. Each case includes a unique identifier, the feature being tested, the exact input expression, and the expected output or behavior.

| 2. Test case ID | 3. Test Item                                              | 4. Input specification | 5. Output Specification | Pass/ Fail |
|-----------------|-----------------------------------------------------------|------------------------|-------------------------|------------|
| TC01            | Basic addition and token recognition                      | 5+2                    | 7                       | Pass       |
| TC02            | Basic subtraction                                         | 6-3                    | 3                       | Pass       |
| TC03            | Multiplication precedence over addition                   | 1+2*3                  | 7                       | Pass       |
| TC04            | Parentheses overriding precedence                         | (4+5) *6               | 54                      | Pass       |
| TC05            | Division operation                                        | 14/7                   | 7                       | Pass       |
| TC06            | Modulo operation                                          | 15 % 8                 | 7                       | Pass       |
| TC07            | Exponentiation operation                                  | 3**3                   | 9                       | Pass       |
| TC08            | Right-to-left exponent associativity                      | 2**3**2                | 512                     | Pass       |
| TC09            | Left-to-right associativity for same-precedence operators | 10-1-4                 | 5                       | Pass       |
| TC10            | Unary negative number                                     | -5+2                   | -3                      | Pass       |

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|      |                                          |               |                                                                       |      |
|------|------------------------------------------|---------------|-----------------------------------------------------------------------|------|
| TC11 | Unary positive number                    | +7*2          | 14                                                                    | Pass |
| TC12 | Floating-point input                     | 1.5+1.5       | 3                                                                     | Pass |
| TC13 | Scientific notation floating-point input | 1e2+5         | 105                                                                   | Pass |
| TC14 | Whitespace handling                      | 8+2           | 10                                                                    | Pass |
| TC15 | Nested parentheses                       | ((2+3)*(4-1)) | 15                                                                    | Pass |
| TC16 | Invalid character detection              | 10+a          | Error message indicating invalid character/token at or near a         | Pass |
| TC17 | Division by zero                         | 1/0           | Error message indicating division by zero                             | Pass |
| TC18 | Mismatched parentheses                   | (2+3          | Error message indicating mismatched parenthesis                       | Pass |
| TC19 | Missing operand                          | 4+            | Error message indicating missing operand or incomplete expression     | Pass |
| TC20 | Incorrect operator sequence              | 5** *2        | Error message indicating invalid operator sequence or missing operand | Pass |

## 6. Environmental needs

### 6.1.1 Hardware

No special hardware is required. The test cases can be executed on any computer capable of compiling and running a C++ program.

### 6.1.2 Software

The program should be compiled using a C++17 compliant compiler such as g++. The test environment should include a command-line terminal for entering expressions and viewing output. No third-party libraries, APIs, or external software frameworks are required beyond the C++ standard library.

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### 6.1.3 Other

The latest version of the project source code should be pulled from the shared GitHub repository before testing. Testers should record actual output and mark each test as pass or fail based on whether the actual output matches the expected output.

## 7. Special procedural requirements

Before executing the test cases, the tester should compile the program successfully and run it from the command line. Each test input should be entered exactly as shown in the table. For error cases, the exact wording of the error message may vary, but the message must clearly identify the correct error type and should not cause the program to crash. After each failed test, the defect should be documented in GitHub or the team's defect log, corrected by the development team, and retested.

## 8. Intercase dependencies

There are no strict intercase dependencies. However, it is recommended that the basic valid-expression tests, such as TC01 through TC07, be executed before the more complex precedence, associativity, nested-parentheses, and error-handling cases. This order helps confirm that the Tokenizer, Parser, Evaluator, and Error Handler are working correctly at a basic level before testing more complex behavior.